

Ali Zulfiqar

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EXPERIENCE

Zikra Infotech LLC

Jun 2025 – present

Software Engineer, MilIOS

Remote

- Built the drawing annotation engine for an agentic millwork estimation platform: deterministic geometry recovery from CAD-exported PDFs, printed dimension and scale-note parsing, and a verified correction loop. Reached exact mark placement (median width error 0.1 in) on production drawings.
- Authored the accompanying preprint, Annotation by Arithmetic ([DOI 10.5281/zenodo.21772948](https://doi.org/10.5281/zenodo.21772948)).
- Own the estimation studio frontend (Next.js) and significant parts of the agent reliability layer: stream recovery, run-state truth, tool-contract hardening across a LangGraph multi-agent pipeline.
- Migrated production model serving between inference providers with a fallback lane and zero behavior change for downstream agents.

Claimlyt

2025 – present

Co-founder

- AI platform automating appeals against insurance claim denials for US healthcare organizations; seven-node LangGraph pipeline.

RESEARCH

Annotation by Arithmetic: Exact Mark Placement on Construction Drawings by Separating Perception from Measurement

2026
Preprint, Zenodo. [doi:10.5281/zenodo.21772948](https://doi.org/10.5281/zenodo.21772948)

- A vision-language model names items and reads printed dimensions; a deterministic engine measures, places, and verifies every mark. Includes a five-class taxonomy of placement failures from instrumented production runs.

SELECTED PROJECTS

- **TURF** – multiplayer territory game on real world maps ([live](#), [code](#)): server-authoritative multiplayer, canvas engine over Leaflet, Playwright smoke suite.
- **ember** – LLM inference engine from scratch in Rust running Qwen2.5 locally ([code](#)).
- **substrate** – a GPT built from first principles in pure JavaScript ([code](#)).
- **blueline** – a browser-based self-driving stack: perception, planning, control ([code](#)).
- **gopher-context** – Go MCP server connecting Claude to any codebase ([code](#)).
- **Zippo** – utility tool, 200+ active users in week one (getzippo.com).

AWARDS AND COMMUNITY

- First place, AI in Civic Innovation hackathon (sachbatao); first place, Aicoo hackathon (front-desk).
- 86k+ followers on Instagram ([@codewithfourtix](#)) and 7k+ on LinkedIn, writing about software and AI.
- Open-source contributions to ScanCode Toolkit.

EDUCATION

FAST-NUCES, Lahore

2024 – 2028 (expected)

BS Computer Science

SKILLS

Languages: TypeScript/JavaScript, Python, Rust, Go, C#, SQL.

Web: React, Next.js (App Router), Node, Express, PostgreSQL, Supabase.

AI/Agents: LangChain, LangGraph, RAG, MCP, agent pipelines, vision model integration, Docker.